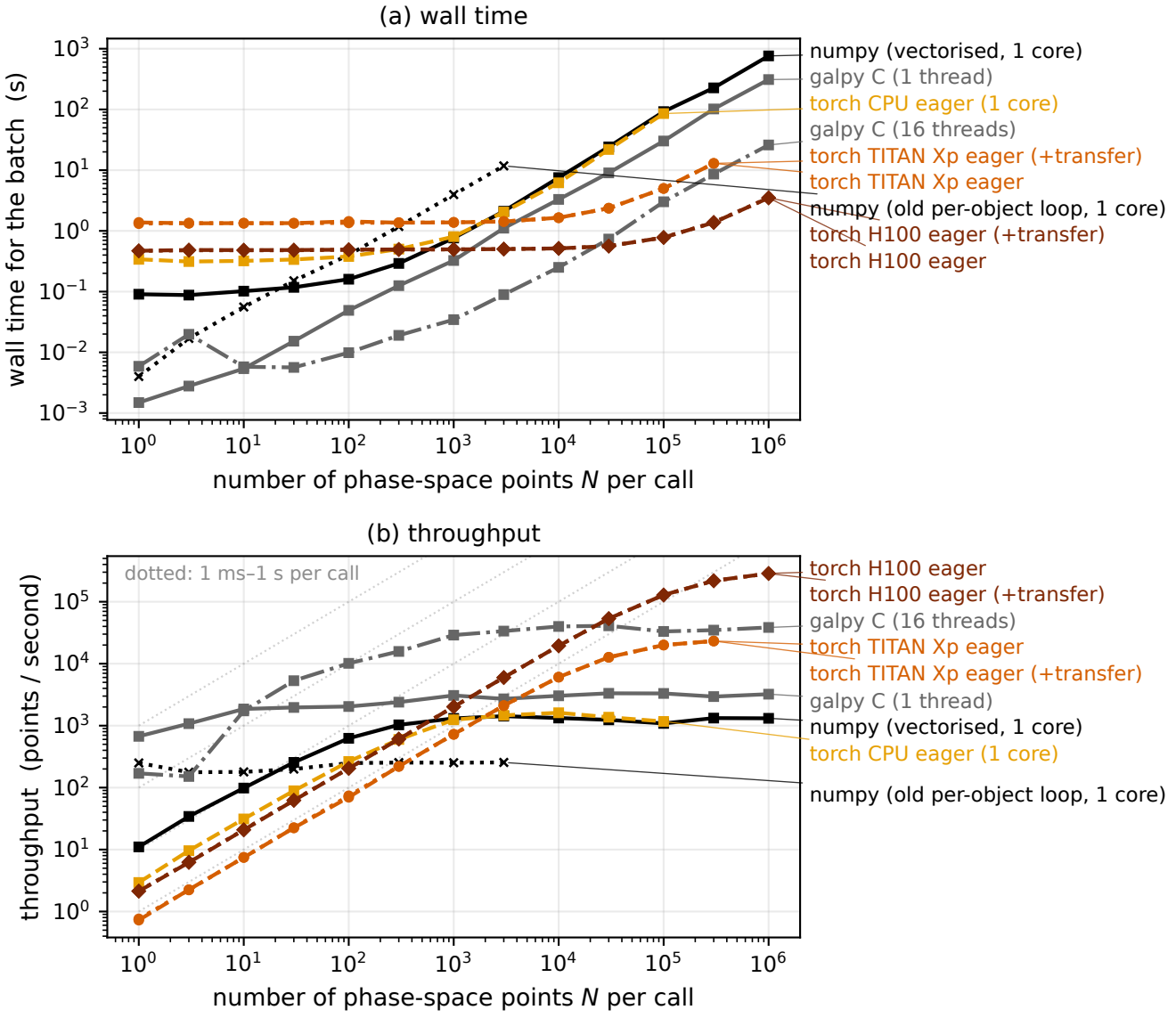
(a) wall time
wall time for the batch (s)
number of phase-space points N per call
numpy (vectorised, 1 core)
galpy C (1 thread)
torch CPU eager (1 core)
galpy C (16 threads)
torch TITAN Xp eager (+transfer)
torch TITAN Xp eager
numpy (old per-object loop, 1 core)
torch H100 eager (+transfer)
torch H100 eager
(b) throughput
dotted: 1 ms–1 s per call
throughput (points / second)
number of phase-space points N per call
torch H100 eager
torch H100 eager (+transfer)
galpy C (16 threads)
torch TITAN Xp eager
torch TITAN Xp eager (+transfer)
galpy C (1 thread)
numpy (vectorised, 1 core)
torch CPU eager (1 core)
numpy (old per-object loop, 1 core)